

The NHDR 2020 reproduction do file: a two-page guide

1. Dos/NHDR2020 reproduction from microdata.do. Stata 16 or later, 27 sections, about five minutes on Stata 18 SE. October 2026.

What the file does

Purpose. The do file reproduces the subnational indices of UNDP Pakistan's National Human Development Report 2020 (HDI, IHDI, CDI, YDI, GDI, GII, LDI, Statistical Annex Tables 1 to 8A) from the raw survey files, extends them with the MPI and a district HDI, and carries every index to 2024-25. It reads 20 national survey rounds (six HIES, four PSLM, six LFS, three PDHS and PMMS 2019), the four provincial MICS6 surveys, and the published tables held in 6. Raw data/Published.

Logic. Each section works in three steps. It reads the raw file and builds an indicator on a stated universe and weight. It applies the index construction, which Sections 5, 16, 17, 18, 23, 24 and 25 first recover by inverting NHDR's own printed tables. It then sets the result beside its target with `nhdr_check`, which writes one line to `Checks.txt`: PASS or FAIL, the value computed, the target and the tolerance. A target tagged [published] is a printed figure from NHDR, PBS, MICS, PDHS or UNDP. A target tagged [reference] is the value an independent Python implementation produced from the same files. The tolerance is half a unit of the last printed digit plus a small margin for floating point.

Reading the code. The capture prefix runs a command and carries on if it fails. The do file uses it only where a failure is expected or harmless: capture program drop clears a program left in memory by an earlier run, capture mkdir tolerates an existing folder, capture confirm file and capture confirm variable test for existence and set `_rc`, and capture noisily { } wraps every chart so that a graphics problem cannot stop the run. Every loop over hard-coded values now carries a comment naming where the values come from: a published table, a DHS or PBS code list, or the independent reference.

Section by section, with the comparison each one makes

Section	What it does, its inputs and the commands that matter	Compared against	Result in the run held in 5. Output CSVs
0	Sets Stata 16 behavior (<code>varabbrev off</code> , every new variable double), finds the package folder with <code>mata direxists()</code> , opens a new run folder and log, unpacks the PDHS zips if needed, confirms every raw file with <code>capture confirm file</code> , opens <code>Checks.txt</code> .	Nothing. A missing file stops the run with its name.	Run folder 8. Stata runs/Run YYYYMMDD HHMMSS.
1	Defines 15 helper programs, each preceded by <code>capture program drop</code> . <code>nhdr_check</code> writes every comparison. <code>nhdr_stack_domains</code> and <code>nhdr_quintile</code> build the 15 domains and quintiles. <code>nhdr_index</code> holds the recovered HDI.	Nothing. Used by every later section.	Not applicable.
2	Reads the published values once with <code>import delimited</code> , <code>asdouble</code> : NHDR Tables 1 and 2A, WDI, model life tables, PBS and MICS benchmarks. Keeps them as <code>tempfiles</code> and <code>scalars</code> .	Nothing. These are the targets.	Not applicable.
3, 4	Literacy 15+ and level-matched net enrollment 5 to 14 from HIES 2018-19 (<code>plst</code> , <code>sec_2ab</code> , <code>sec_12ce</code>). Stacks domains, cuts consumption quintiles within each domain, collapses with person weights.	Table 2A, 15 domains x 6 rows.	Pakistan 57.40 against 57.4 and 37.00 against 37.0. Worst domain gap 0.05. Quintile rows within 4.85 and 5.06 points.
5	Pushes the Table 2A inputs through the recovered construction: education 2/3 literacy + 1/3 enrollment, health 25 to 90 years, income 100 to 100,000 PPP dollars, geometric mean.	Table 1, 120 indices.	Worst error 0.0011, the rounding of the inputs. No UNDP report uses these goalposts.
6	Builds Pakistan's 2017 HDI on the current UNDP method from its four published inputs.	UNDP 0.562.	0.5618.
7	Recovers NHDR's life expectancy as $25 + 65 \times$ the printed health index.	Table 2A life expectancy, 30 values.	30 of 30 inside the rounding band, worst 0.07 years.
8	Under-five mortality from 401,088 MICS6 birth records (four provinces) by synthetic cohort (<code>nhdr_q5</code>), then life expectancy (Coale-Demeny West).	MICS6 reports, PDHS 2017-18, NHDR.	Punjab 67.3 against 67, Sindh 45.2 against 46. MICS6 life expectancy runs 2.9 to 7.0 years above PDHS.
8A	The same from PDHS 2006-07, 2012-13 and 2017-18 with a port of the DHS Program's own code (<code>nhdr_q5_dhs</code>). Stunting, wasting, wealth gradient, Figure 5.19.	PDHS reports and Table 4A.	q5 94.2, 89.0, 74.0 against 94, 89, 74. Male wasting is the one failed check (1.29 points).
9	Allocates 2018-19 value added across provinces on Pasha's scheme, 22 sub-sectors, with allocators from LFS, HIES, OCAC, PBS electricity and CMI.	Pasha 2021, KP BOS, PIDE.	95.0 percent of GDP allocated. Punjab 57.6, Sindh 27.6, KP 10.4, Balochistan 4.4 percent.
10	Tests what NHDR's income column is: provincial pattern, ranking variable, level.	Table 2A income column.	Level is World Bank GNI PPP (within 0.45 percent). Quintiles are consumption quintiles (gap 0.046 against 0.288). No regional product estimate matches the pattern.
11	Reads HIES 2024-25 (30,123 households) and builds consumption with the PBS rule.	PBS HIES 2024-25 Key Findings Report.	Consumption within 1 percent, literacy 10+ 63.06 against 63.

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12, 13	HDI 2024-25 as the surveys report it, then on one instrument and one price basis, and the log decomposition of the change.	Table 1 2018-19.	Pakistan 0.588 against 0.570. Education 1.43, health 1.01, income 0.80 percent. Pace 3.1 points per 1,000 a year.
14, 15	District education from PSLM 2019-20 with cluster standard errors, then the district and divisional HDI with a district, division, province fallback for health.	PBS District Level Report.	126 districts, Karachi East 0.717 to Dera Bugti 0.358. 65 low, 58 medium, 3 high. Median SE 2.9 points.
16	IHDI: Atkinson index over the five quintiles, first from Table 2A, then from the microdata (MICS6 or PDHS health), then 2024-25.	Table 3, Figure 2.19.	Pakistan 0.534 against 0.534, loss 6.28 against 6.26. Income carries 75 percent of the inequality.
17	GDI: recovers the construction (income ceiling 75,000, not 100,000), splits income by sex with the UNDP method on LFS.	Tables 6, 6A.	0.750 and 0.777 reproduced. Female income 1,799 against NHDR 1,673: NHDR's split is not recoverable.
18	GII: recovers NHDR's variant (no-care rate and early marriage replace MMR and adolescent births), reproduces the inputs from HIES and LFS.	Table 7, PBS LFS tables.	0.548 reproduced. NHDR's 2018-19 participation is LFS 2017-18. Early marriage 12.3 against NHDR 20.0.
19, 21	MPI 2019-20 and 2014-15 by Alkire-Foster from PSLM, 14 indicators, and the harmonized change.	MPI Report 2019-20.	0.1455 against 0.146, H 30.4 against 30.5, 125 of 126 districts inside the interval. Change -0.022 against -0.021.
20	The 2006-07 column from HIES 2005-06, and the HDI with every dimension from the microdata (hdi_e, hdi_eh, hdi_ehi) and with Pasha's income.	Tables 2A, 3, 6A, 7A.	Literacy 50.66 against 50.7, enrollment 34.71 against 34.7, IHDI 0.493 against 0.491.
22	Reads LFS 2006-07, 2012-13, 2014-15, 2017-18 and 2024-25 on one set of definitions.	Table 7A.	LFS 2017-18 reproduces NHDR's 2018-19 column within 0.06.
23	CDI from HIES and LFS with PDHS health inputs, and the PSLM 2006-07 and 2008-09 test.	Tables 4, 4A.	Worst gap 0.025, Pakistan 0.007. CDI 2024-25 0.572.
24	YDI from LFS and PMMS 2019. Shows the survival goalposts are misprinted (0.9770 should read 0.9970).	Tables 5, 5A.	Pakistan within 0.003 of 0.605, worst 0.017. YDI 2024-25 0.681.
25	LDI from LFS with WDI GDP and Pasha's shares. Confirms the geometric mean.	Tables 8, 8A.	Worst gap 0.034, or 0.010 with NHDR's labor share. LDI 2024-25 0.422.
26A, 26	Builds the four-year national series (2006-07, 2012-13, 2015-16, 2018-19) that figures.py plots, then draws 20 Stata figures and 8 maps with graph bar, graph hbar and twoway inside capture noisily.	The published figures.	29 of 29 figures and maps written.
27	Closes Checks.txt, draws four extra charts, writes the results workbook (one sheet per table, putexcel headlines) and prints the summary.	All checks.	298 of 299 checks pass.

Where the plots come from

Stata charts. Sections 8A.5, 26 and 27.1 draw the Stata charts and write the data behind each figure (Figure *.csv). Build outputs.py copies the charts to 4. Plots/4. Stata charts/Charts and the data to 5. Output CSVs/Figure data.

NHDR-style figures. The figures in 4. Plots/2. Figures reproduced NHDR style are not drawn by the do file. 7. Python tools/figures.py draws them in the report's layout and typeface from the result tables the do file writes: HDI all microdata.csv and IHDI all microdata.csv (Section 20.6), the Series files of Section 26A, and the CDI, YDI, GDI, GII and LDI tables of Sections 17 to 25. juxtapose.py then sets each beside the figure cropped from the report (folders 1 and 3). The two reproduced sets do not always plot the same series. The NHDR-style figures plot the all-microdata four-year series, while several Stata charts plot the 2018-19 and 2024-25 variants.

What the comparison shows

Constructions. Every published index is reproduced from NHDR's own inputs within the rounding of those inputs. Each construction is now known exactly. The HDI uses goalposts no UNDP report uses (health 25 to 90 years, income 100 to 100,000 PPP dollars), and the GDI uses a different income ceiling (75,000) in the same report. The YDI survival goalposts are misprinted. NHDR documents none of it.

Inputs. Education reproduces from HIES to the printed decimal. The health result depends on the survey. MICS6 gives life expectancy 2.9 to 7.0 years above PDHS, which reorders the provinces. Income reproduces least well. Its level is World Bank GNI per head, its distribution is survey consumption scaled up about threefold, and its provincial pattern matches no published estimate of regional product, Pasha's included. NHDR's labor indicators labeled 2018-19 are LFS 2017-18, and its early-marriage rate (20.0 percent) is reproduced by no survey on disk (12.3 to 12.6 percent).

What remains open. Male wasting in Table 4A cannot be reproduced from PDHS 2017-18 and is the one failed check. The GDI income split, the GII early-marriage column and the provincial income pattern need confirmation from UNDP Pakistan before the 2024-25 indices are published as a continuation of the NHDR series.